Experiment - 14

Aim - Write a code using RAW sockets to implement packet sniffing.

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protocol_name = ""
        if protocol == 1:
            protocol = name = 'ICMP'
        elif protocol = 6:
            protocol = name = "TCP"
        else
            protocol_name = "Unknown Protocol"
        print(f"Protocol: {protocol_name}")
        print(f"Source IP: {src_ip}")
        print("_" * 50
sniff(iface="wifi", prn=packet_callback, filter="ip"
      store=0)
```

Output:

Protocol : TCP
Source IP : 57.155.141.117
Destination IP : 192.168.1.6

Protocol : UDP
Source IP : 57.155.141.117
Destination IP : 192.168.1.1

Result:

Thus, the program has been executed.